Machine Learning: Introduction

Elsa Ferreira Gomes

2025/2026

Machine Learning

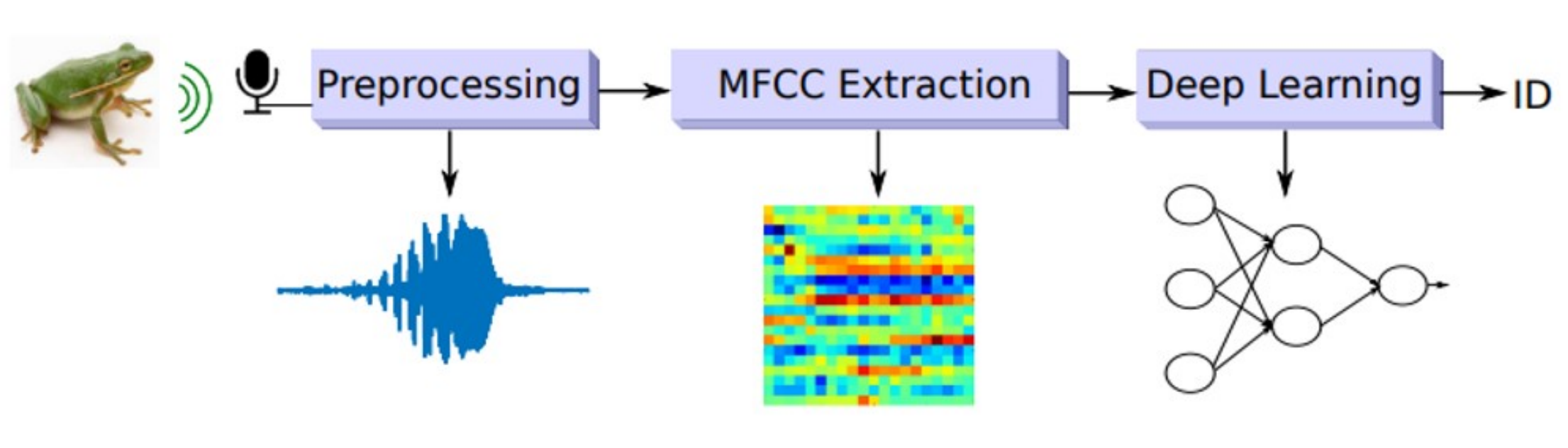
Problem examples

Classification using audio data: heart sounds and other signals

Machine Learning

Problem examples

Classification anuran sounds

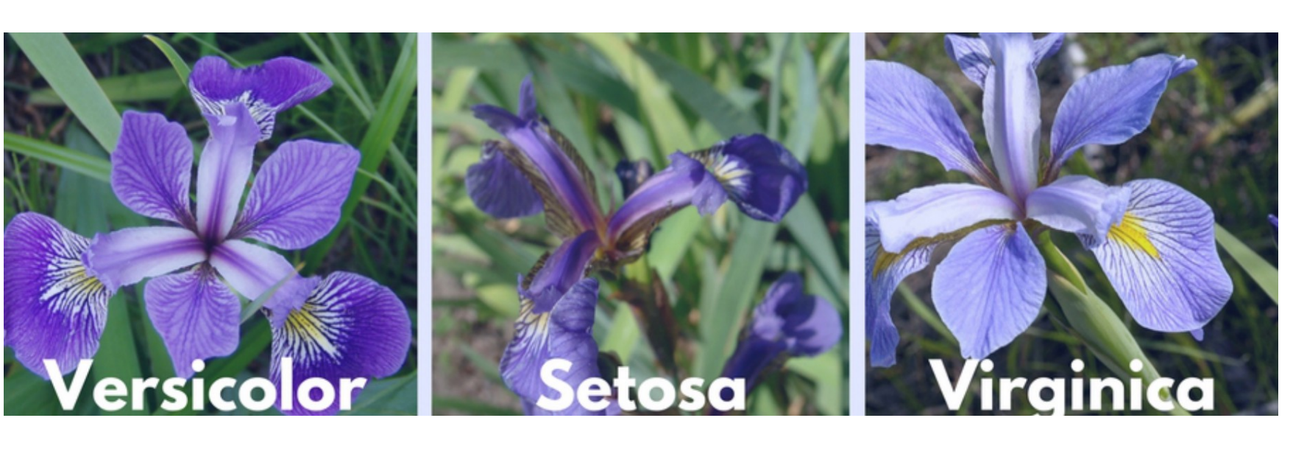


Machine Learning

Problem examples

Flower classification

Classify types of Iris flowers (setosa, versicolor, or virginica). How would you do it?



Machine Learning

Problem examples:

- Spotify Popularity Prediction
- Predict House Prices
- Natural Language Processing with Disaster Tweets
- Email spam detection system
- Histopathologic Oral Cancer Detection
- Identify numbers in a handwritten zip code
- Face recognition
- .....

Machine Learning

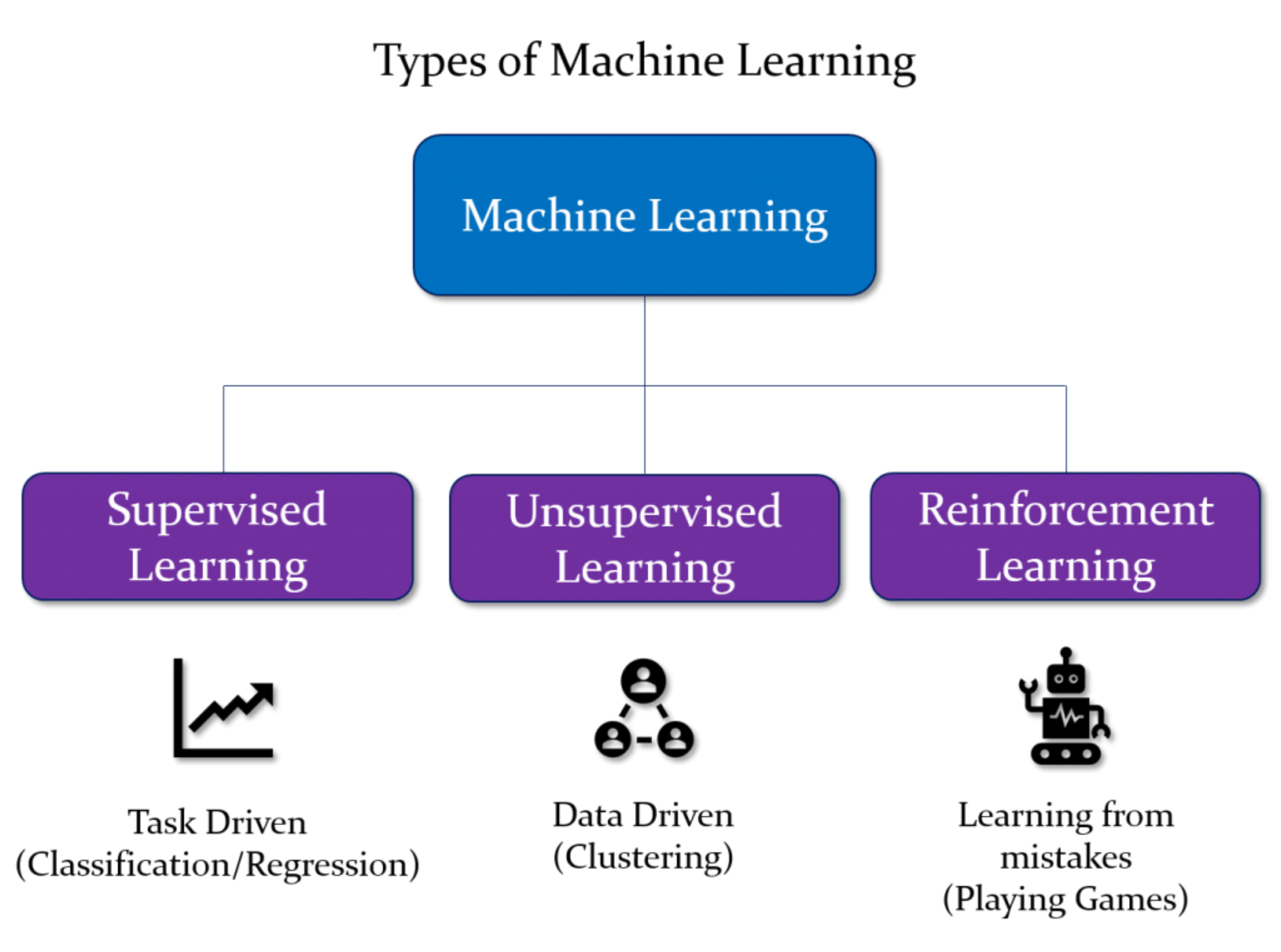
"A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in *T*, as measured by *P*, improves with experience *E*."

Tom Mitchell

Example: A spam filter learns if, after seeing more labeled emails (experience E), it gets better at classifying new emails (task T) according to its accuracy (performance measure P).

- Task (T): Classifying emails as spam or not spam.
- Performance Measure (P): The percentage of emails correctly classified
- Experience (E): A dataset of labeled emails (spam / not spam)

Machine Learning



Types of Machine Learning

Machine Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

Task Driven (Classification/Regression)

Data Driven (Clustering)

Learning from mistakes (Playing Games)

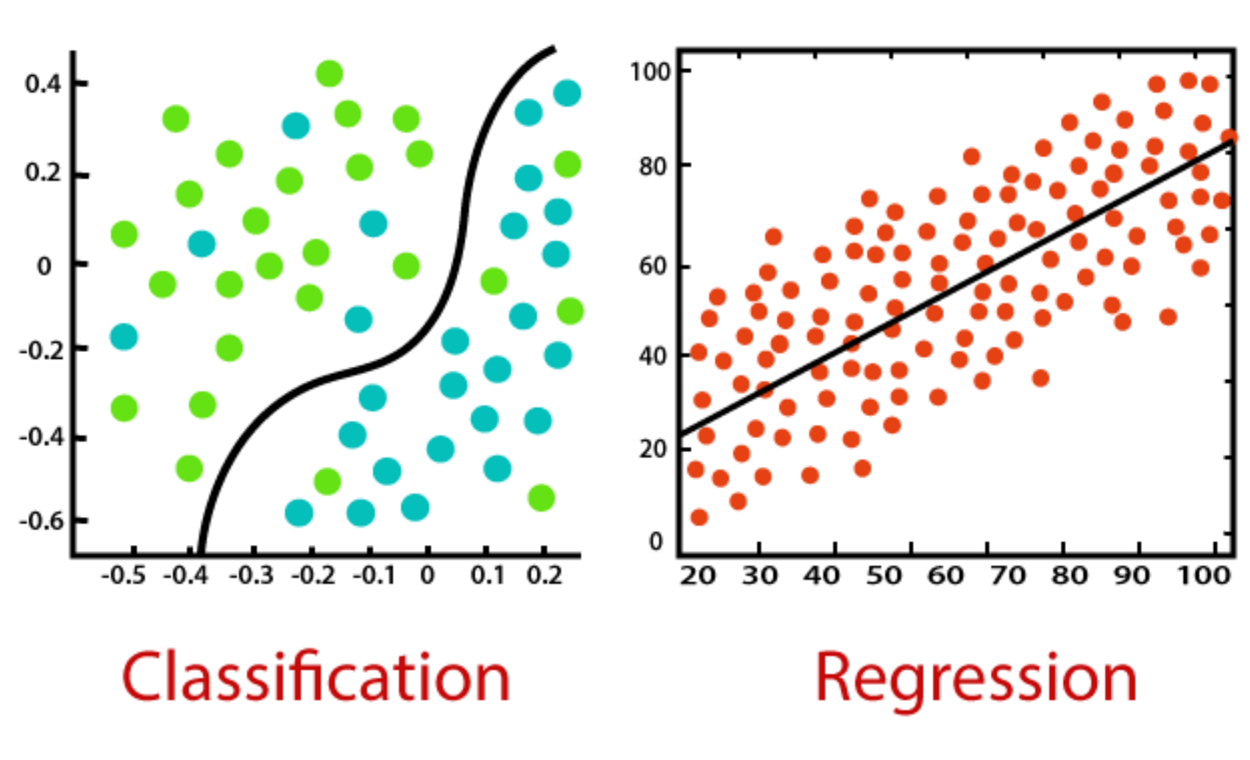
Types of machine learning

- **Supervised Learning**: learn a *model* f from *labeled data* (X, y) (ground truth)
  - Given a new input *X*, predict the right output *y*
  - Given examples of stars and galaxies, identify new objects in the sky
- **Unsupervised Learning**: explore the structure of the data (X) to extract meaningful information
  - Given inputs *X*, find which ones are special, similar, anomalous, ...
- **Reinforcement Learning**: develop an agent that improves its performance based on interactions with the environment

Note: Practical ML systems can combine many types in one system.

Supervised Learning Problem

- Outcome measurement Y (also called dependent variable, response, target).
- Vector of p predictor measurements X (or inputs, regressors, covariates, features, independent variables).
- In the **regression problem**, Y is quantitative (e.g price, blood pressure).
- In the **classification problem**, Y takes values in a finite, unordered set (e.g survived/died, digit 0-9, cancer class of tissue sample).
- The training data $(x_1, y_1), \ldots, (x_N, y_N)$ are observations (examples, instances) of these measurements.



Supervised Learning Problem

Objectives

On the basis of the training data we would like to:

- Accurately predict **unseen test cases**.
- Understand which inputs (and how) affect the outcome.
- Assess the quality of our predictions and inferences.

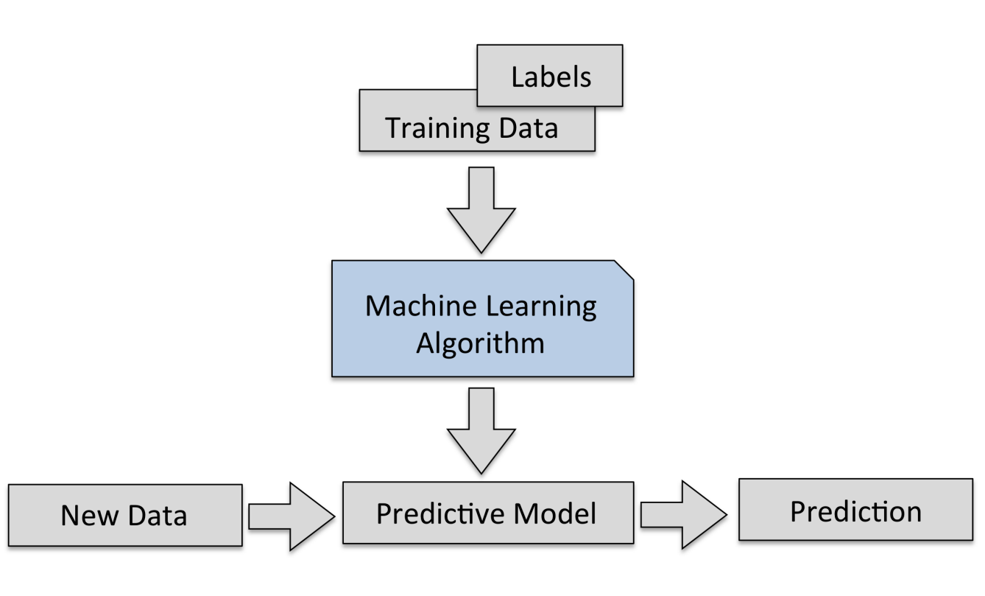
Supervised Learning Problem

The aim

Our goal is to find a **useful approximation** $\hat{f}(x)$ of $f(x)$

- $f(x)$ is the function that generates the phenomenon it is **unknown**
- What is the best **approximation**?
- How do we measure the **goodness** of an approximation?

- How do we **find the best approximation** of $f(x)$ (or at least a very good one)?



Labels

Training Data

Machine Learning Algorithm

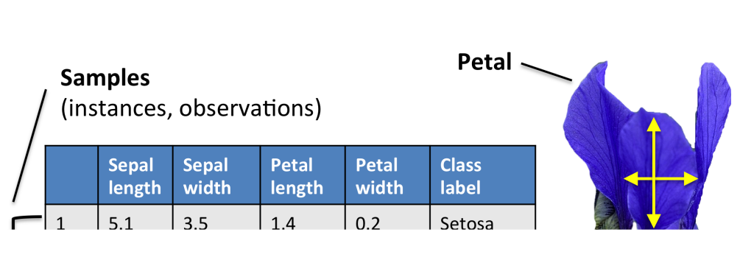
New Data

Predictive Model

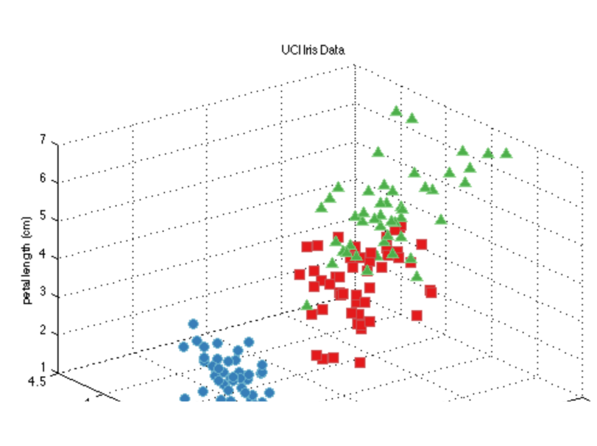
Prediction

Representation: input features and labels

- We can manually define a number of input features (variables), e.g. length and width of leaves
- Every 'example' is a point in a (possibly high-dimensional) space


Samples
(instances, observations)
Petal

	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa



Classification

- Predict a *class label* (category), discrete and unordered
  - Can be *binary* (e.g. spam/not spam) or *multi-class* (e.g. letter recognition)
  - Many classifiers can return a *confidence* per class
- The predictions of the model yield a *decision boundary* separating the classes

Regression

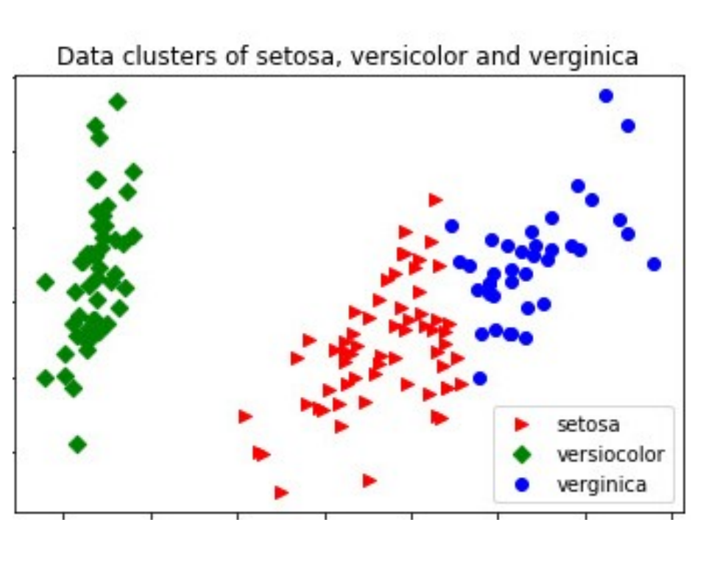
- Predict a continuous value, e.g. temperature
  - Target variable is numeric
  - Some algorithms can return a *confidence interval*
- Find the relationship between predictors and the target.

Unsupervised Learning Problem

- No outcome variable
  - just a set of predictors (features) measured on a set of samples.
- Explore the structure of the data to extract information
- Different from supervised learning,
  - but can be useful as a pre-processing step for supervised learning.

Clustering

- Organize information into meaningful subgroups (clusters)
- Objects in cluster share certain degree of similarity (and dissimilarity to other clusters)

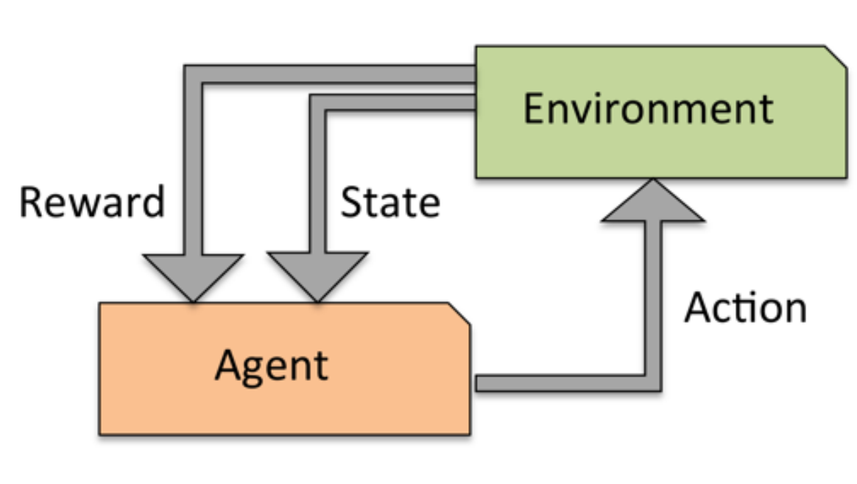


Reinforcement learning

- Develop an agent that improves its performance based on interactions with the environment
- Search a space of actions and states
- *Reward function* defines how well the actions works
- Learn a series of actions (policy) that maximizes reward through exploration

- Example:

  - Games like Chess, Go (AlphaGo/AlphaZero), Atari ...
  - Autonomous Vehicles: Self-driving cars can use RL to optimize driving strategies in simulations,logistics (truck platooning)
  - Robotics: manipulation, locomotion (like quadrupeds or humanoids learn to walk, run...), drone navigation (like learn obstacle avoidance)...



Machine Learning

Example: Predicting Used Car Prices

- We have observations (experience)
- We assume there is a function that outputs the position given the (car) age

$$price = f(car.age)$$

```
In [1]: import matplotlib.pyplot as plt
        import numpy as np
        from sklearn.linear_model import LinearRegression
        import pandas as pd

        # create a dataframe from scratch using a dictionary
        used_cars = pd.DataFrame({
            'car_age': [4, 4, 5,  5,  7,  7, 8, 9, 10, 11, 12],
            'price': [6300, 5800, 5700, 4500, 4500, 4200, 4100, 3100, 2100, 2500, 2200]
        })

        used_cars
```

Out[1]:

	car_age	price
0	4	6300
1	4	5800
2	5	5700
3	5	4500
4	7	4500
5	7	4200
6	8	4100
7	9	3100
8	10	2100
9	11	2500
10	12	2200

Python libraries

Pandas

Pandas is used to analyze data.

Numpy

NumPy is the fundamental package for scientific computing in Python used for working with arrays (domain of linear algebra, fourier transform, and matrices)

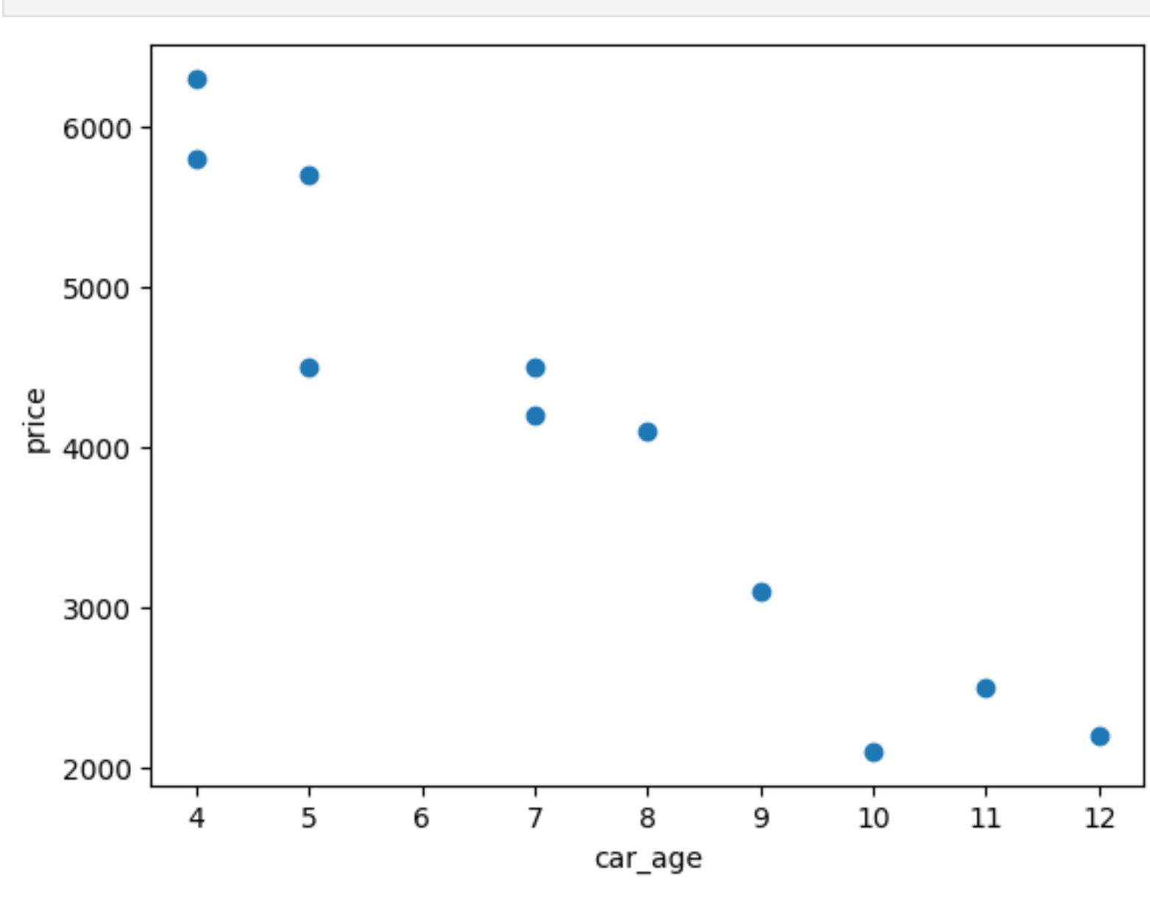
Sklearn

Machine Learning in Python (is simple and efficient tools for predictive data analysis)

Matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python

```
In [2]: ax=plt.axes()
        ax.scatter(used_cars.car_age,used_cars.price)
        ax.set(xlabel='car_age')
        ax.set(ylabel='price'); # this semicolon supresses the output of this command. Try without
```



Machine Learning

The goal

- Find a **useful approximation** $\hat{f}(x)$ of $f(x)$

- We need to know how we:

  - **find** the best **approximation**
  - **measure** the **goodness**
  - **find the best approximation**

Linear model

Find a linear function

- We can use **linear functions**

$$\hat{y} = \hat{f}(x) = \beta_0 + \beta_1 x$$

- Now we have to look for one linear function that suits us
  - All we have, all we know, is our **data**
  - There are **many** different ways to do that
  - **One** is the **least squares** method

We can use a linear model to aproximate the function

- Linear Regression

```
In [3]: x=np.array(used_cars.car_age)
        X=x[:, np.newaxis] # newaxis gives X the shape of a matrix. Try without
        y=np.array(used_cars.price)

        model = LinearRegression(fit_intercept=True)
        model.fit(X,y)

        # Pretty print model
        beta1=model.coef_[0]
        beta0=model.intercept_

        print('price =',beta0,'+',beta1,'*car_age')
```

```
price = 7836.258660508083 + -502.4249422632795 *car_age
```

Approximating the function

- Here, the approximated function $\hat{f}(x)$ is:

$$price = \hat{f}(car.age) = 7836.25 - 502.42 \times car.age$$

Hint: The coefficient estimates for Ordinary Least Squares rely on the independence of the features.

How does the model approximate the true function

Calculate the residuals

```
In [4]: ypred=model.predict(X)
        residuals=y-ypred
        residuals
```

```
Out[4]: array([ 473.44110855,  -26.55889145,  375.86605081, -824.13394919,
                180.71593533, -119.28406467,  283.1408776 , -214.43418014,
               -712.00923788,  190.41570439,  392.84064665])
```

```
In [5]: # Residual Sum of Squares

        sum(residuals**2)
```

```
Out[5]: 1915900.692840648
```

```
In [6]: quartiles = np.percentile(residuals, [25, 50, 75])
        maxr=max(residuals)
        minr=min(residuals)
        # Printing with format (%). We define number of digits to print in floats (.3f)
        print('Residuals summary:')
        print('min = %.3f' % minr)
        for i in range(3):
            print('Q%i = %.3f' % (i+1, quartiles[i]))
        print('max = %.3f' % maxr)
```

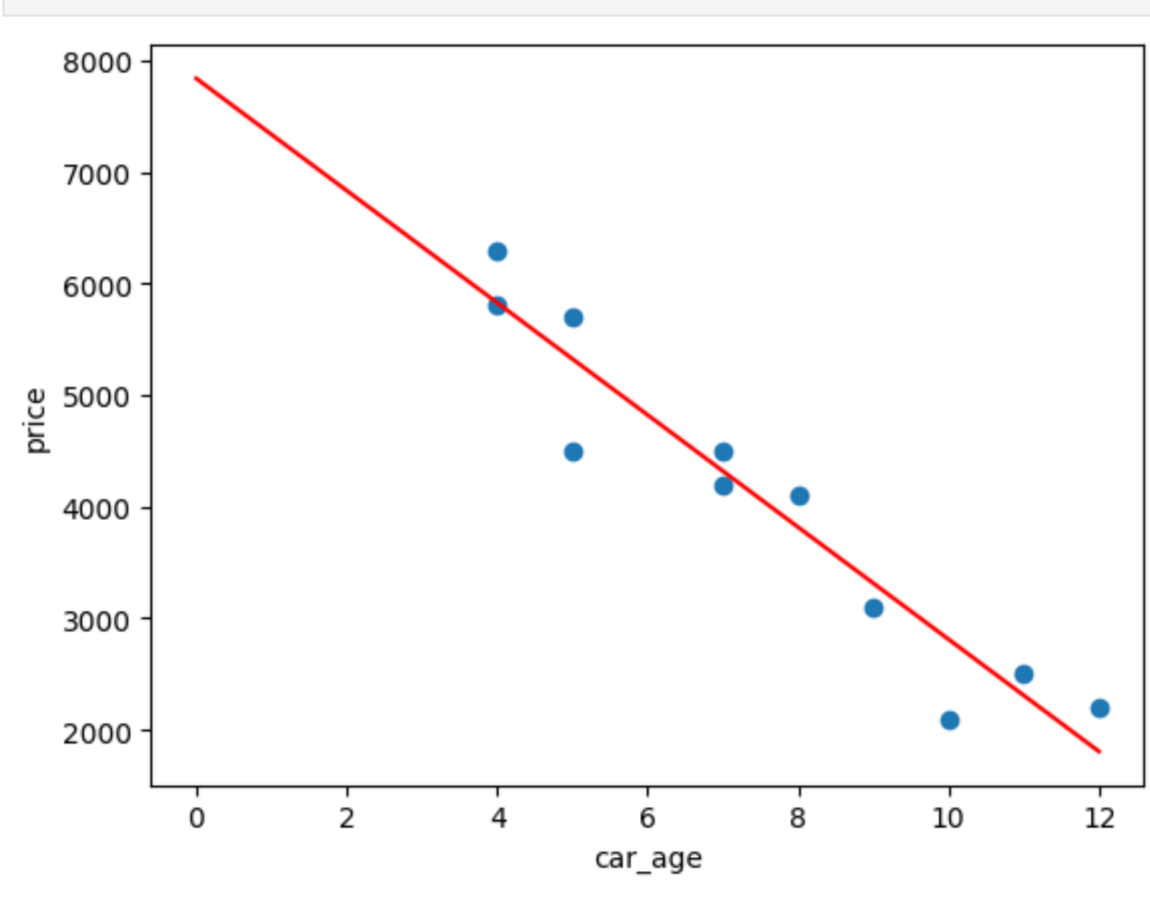
```
Residuals summary:
min = -824.134
Q1 = -166.859
Q2 = 180.716
Q3 = 329.503
max = 473.441
```

```
In [7]: # plot the regression line against the points

        xfit = np.array([0,max(x)])

        # newaxis gives xfit the shape of a matrix
        yfit = model.predict(xfit[:,np.newaxis])

        ax=plt.axes()
        ax.scatter(x,y)
        ax.set(xlabel='car_age')
        ax.set(ylabel='price');
        ax.plot(xfit, yfit,color='red');
```



```
In [8]: # Did we find a useful approximation?
        # Let's try some test cases

        xtest=np.array([0.1,1,5,10,100])
        Xtest=xtest[:,np.newaxis] # Xtest is a matrix now
        ypred = model.predict(Xtest)
        ypred
```

```
Out[8]: array([  7786.01616628,   7333.83371824,   5324.13394919,   2812.00923788,
               -42406.23556582])
```

```
In [9]: # The output is a float, so we should round to an integer
        list(map(lambda predpos: int(np.round(predpos,0)),ypred))
```

```
Out[9]: [7786, 7334, 5324, 2812, -42406]
```

```
In [11]: # How good is the model on average?
         # We will measure R2 of the model

         print('R2 of the linear model is %.3f' % model.score(X,y))
```

```
R2 of the linear model is 0.912
```

```
In [ ]:
```